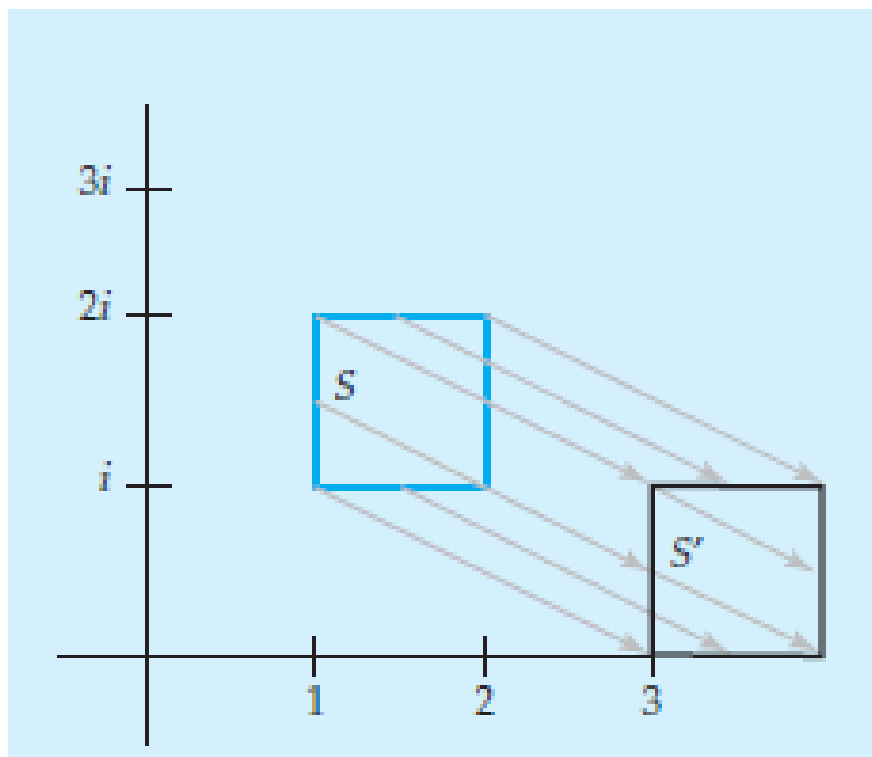




Complex Integration

<ftp://10.220.20.26/academic/mce/maka>

Definite Integral

$$\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a). \quad (1)$$

Steps Leading to the Definition of the Definite Integral

1. Let f be a function of a single variable x defined at all points in a closed interval $[a, b]$.
2. Let P be a partition:

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$$

of $[a, b]$ into n subintervals $[x_{k-1}, x_k]$ of length $\Delta x_k = x_k - x_{k-1}$.
See Figure 5.1.

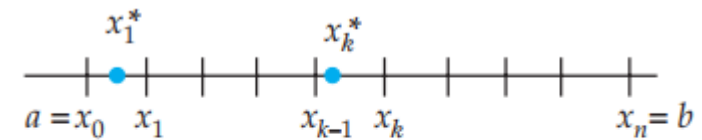


Figure 5.1 Partition of $[a, b]$ with x_k^* in each subinterval $[x_{k-1}, x_k]$

Definite Integral

3. Let $\|P\|$ be the **norm** of the partition P of $[a, b]$, that is, the **length of the longest subinterval**.
4. Choose a number x_k^* in each subinterval $[x_{k-1}, x_k]$ of $[a, b]$. See Figure 5.1.
5. Form n products $f(x_k^*)\Delta x_k$, $k = 1, 2, \dots, n$, and then sum these products:

$$\sum_{k=1}^n f(x_k^*) \Delta x_k.$$

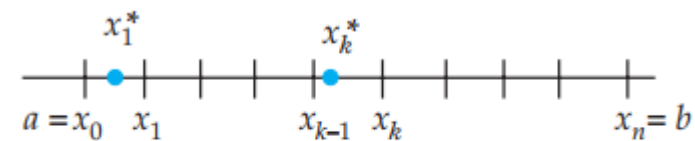


Figure 5.1 Partition of $[a, b]$ with x_k^* in each subinterval $[x_{k-1}, x_k]$

Definition 5.1 Definite Integral

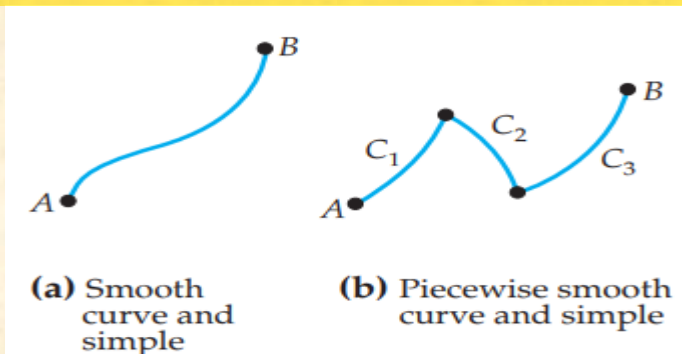
The definite integral of f on $[a, b]$ is

$$\int_a^b f(x) dx = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(x_k^*) \Delta x_k. \quad (2)$$

Terminology

Suppose a curve C in the plane is parametrized by a set of equations $x = x(t)$, $y = y(t)$, $a \leq t \leq b$, where $x(t)$ and $y(t)$ are continuous real functions. Let the **initial** and **terminal points** of C , that is, $(x(a), y(a))$ and $(x(b), y(b))$, be denoted by the symbols A and B , respectively. We say that:

- (i) C is a **smooth curve** if x' and y' are continuous on the closed interval $[a, b]$ and not simultaneously zero on the open interval (a, b) .
- (ii) C is a **piecewise smooth curve** if it consists of a finite number of smooth curves $C_1, C_2, \dots, C_n$ joined end to end, that is, the terminal point of one curve C_k coinciding with the initial point of the next curve C_{k+1} .



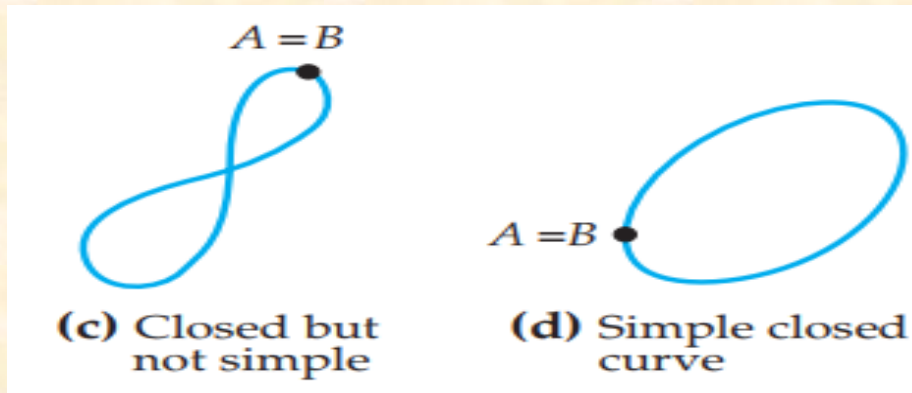
Terminology

Suppose a curve C in the plane is parametrized by a set of equations $x = x(t)$, $y = y(t)$, $a \leq t \leq b$, where $x(t)$ and $y(t)$ are continuous real functions. Let the **initial** and **terminal points** of C , that is, $(x(a), y(a))$ and $(x(b), y(b))$, be denoted by the symbols A and B , respectively. We say that:

(iii) C is a **simple curve** if the curve C does not cross itself except possibly at $t = a$ and $t = b$.

(iv) C is a **closed curve** if $A = B$.

(v) C is a **simple closed curve** if the curve C does not cross itself and $A = B$; that is, C is simple and closed.



Steps Leading to the Definition of Line Integrals

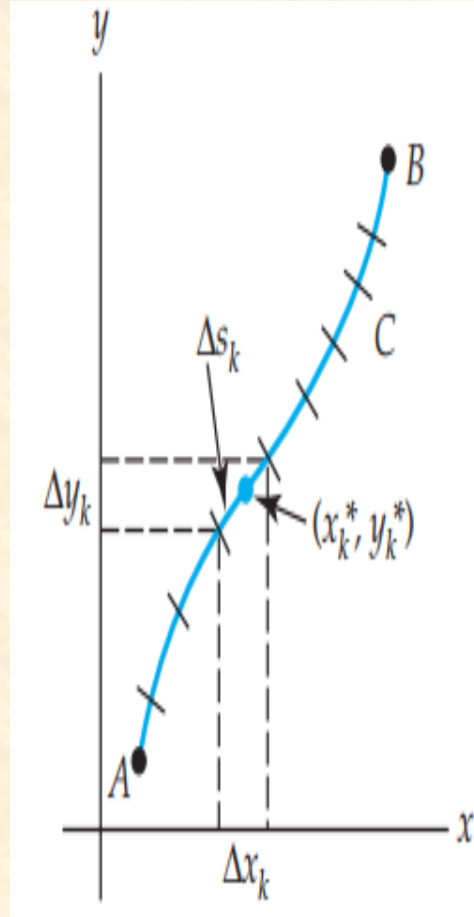
1. Let G be a function of two real variables x and y defined at all points on a smooth curve C that lies in some region of the xy -plane. Let C be defined by the parametrization $x = x(t)$, $y = y(t)$, $a \leq t \leq b$.

2. Let P be a partition of the parameter interval $[a, b]$ into n subintervals $[t_{k-1}, t_k]$ of length $\Delta t_k = t_k - t_{k-1}$:

$$a = t_0 < t_1 < t_2 < \cdots < t_{n-1} < t_n = b.$$

The partition P induces a partition of the curve C into n subarcs of length Δs_k . Let the projection of each subarc onto the x - and y -axes have lengths

9/23/2024 Δx_k and Δy_k , respectively. See Figure 5.3.



Steps Leading to the Definition of Line Integrals

3. Let $\|p\|$ be the norm of the partition P of $[a, b]$, that is, the length of the longest subinterval.
4. Choose a point (x_k^*, y_k^*) on each subarc of C . See Figure 5.3.
5. Form n products $G(x_k^*, y_k^*)\Delta x_k$, $G(x_k^*, y_k^*)\Delta y_k$, $G(x_k^*, y_k^*)\Delta s_k$, $k = 1, 2, \dots, n$, and then sum these products

$$\sum_{k=1}^n G(x_k^*, y_k^*)\Delta x_k$$
$$\sum_{k=1}^n G(x_k^*, y_k^*)\Delta y_k$$
$$\sum_{k=1}^n G(x_k^*, y_k^*)\Delta s_k$$

Line Integrals in the plane

(i) The line integral of G along C with respect to x is

$$\int_C G(x, y) dx = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n G(x_k^*, y_k^*) \Delta x_k. \quad (3)$$

(ii) The line integral of G along C with respect to y is

$$\int_C G(x, y) dy = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n G(x_k^*, y_k^*) \Delta y_k. \quad (4)$$

(iii) The line integral of G along C with respect to arc length s is

$$\int_C G(x, y) ds = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n G(x_k^*, y_k^*) \Delta s_k. \quad (5)$$

Line Integral

Example 1: Evaluate (a) $\int_C xy^2 dx$, (b) $\int_C xy^2 dy$, and (c) $\int_C xy^2 ds$, where the path of integration C is the quarter circle defined by $x = 4 \cos t$, $y = 4 \sin t$, $0 \leq t \leq \pi/2$.

Solution The path C of integration is shown in color in Figure 5.4. In each of the three given line integrals, x is replaced by $4 \cos t$ and y is replaced by $4 \sin t$.

(a) Since $dx = -4 \sin t dt$, we have from (6):

$$\begin{aligned}\int_C xy^2 dx &= \int_0^{\pi/2} (4 \cos t) (4 \sin t)^2 (-4 \sin t dt) \\ &= -256 \int_0^{\pi/2} \sin^3 t \cos t dt = -256 \left[\frac{1}{4} \sin^4 t \right]_0^{\pi/2} = -64.\end{aligned}$$

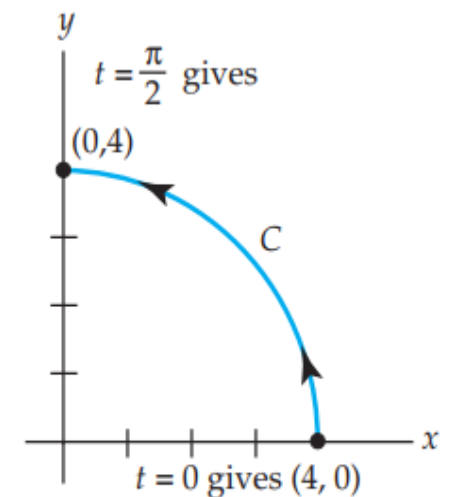


Figure 5.4 Path C of integration

Line Integral

Example 1: Evaluate **(a)** $\int_C xy^2 dx$, **(b)** $\int_C xy^2 dy$, and **(c)** $\int_C xy^2 ds$, where the path of integration C is the quarter circle defined by $x = 4 \cos t$, $y = 4 \sin t$, $0 \leq t \leq \pi/2$.

(b) Since $dy = 4 \cos t dt$, we have from (7):

$$\begin{aligned}\int_C xy^2 dy &= \int_0^{\pi/2} (4 \cos t) (4 \sin t)^2 (4 \cos t dt) \\&= 256 \int_0^{\pi/2} \sin^2 t \cos^2 t dt \\&= 256 \int_0^{\pi/2} \frac{1}{4} \sin^2 2t dt \\&= 64 \int_0^{\pi/2} \frac{1}{2} (1 - \cos 4t) dt = 32 \left[t - \frac{1}{4} \sin 4t \right]_0^{\pi/2} = 16\pi.\end{aligned}$$

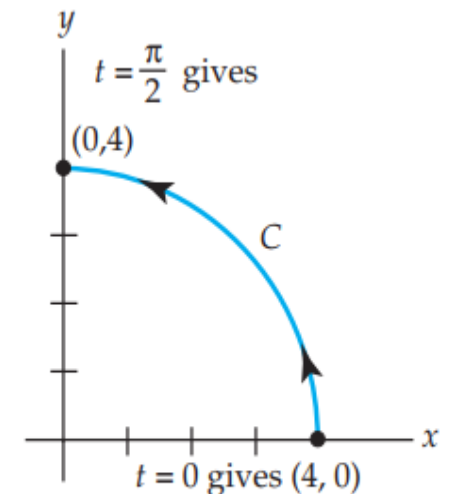


Figure 5.4 Path C of integration

Line Integral

Example 1: Evaluate (a) $\int_C xy^2 dx$, (b) $\int_C xy^2 dy$, and (c) $\int_C xy^2 ds$, where the path of integration C is the quarter circle defined by $x = 4 \cos t$, $y = 4 \sin t$, $0 \leq t \leq \pi/2$.

(c) Since $ds = \sqrt{16(\sin^2 t + \cos^2 t)} dt = 4 dt$, it follows from (8):

$$\begin{aligned}\int_C xy^2 ds &= \int_0^{\pi/2} (4 \cos t) (4 \sin t)^2 (4 dt) \\ &= 256 \int_0^{\pi/2} \sin^2 t \cos t dt = 256 \left[\frac{1}{3} \sin^3 t \right]_0^{\pi/2} = \frac{256}{3}.\end{aligned}$$

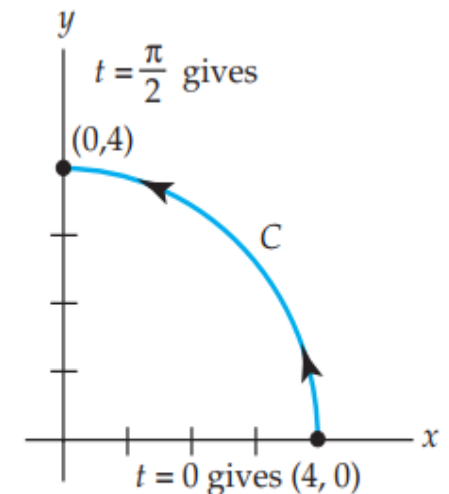


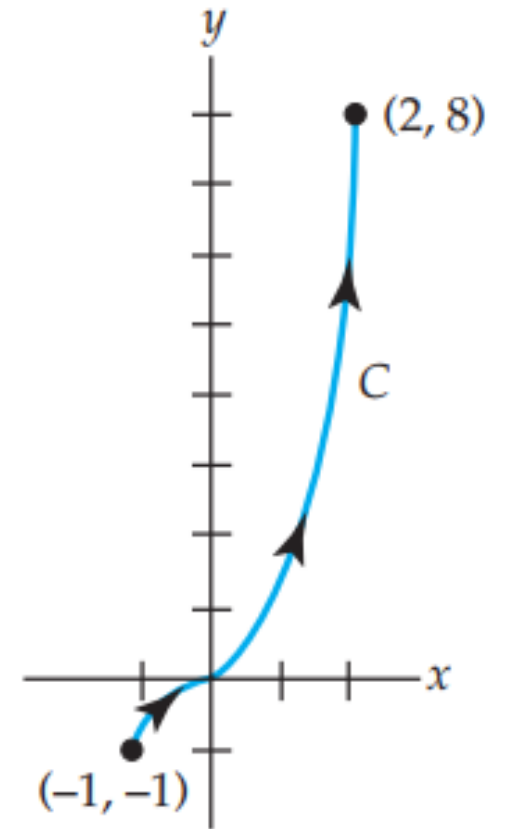
Figure 5.4 Path C of integration

Line Integral

Example 2: Evaluate $\int_C xy dx + x^2 dy$, where the path of integration C is $y = x^3$, $-1 \leq x \leq 2$.

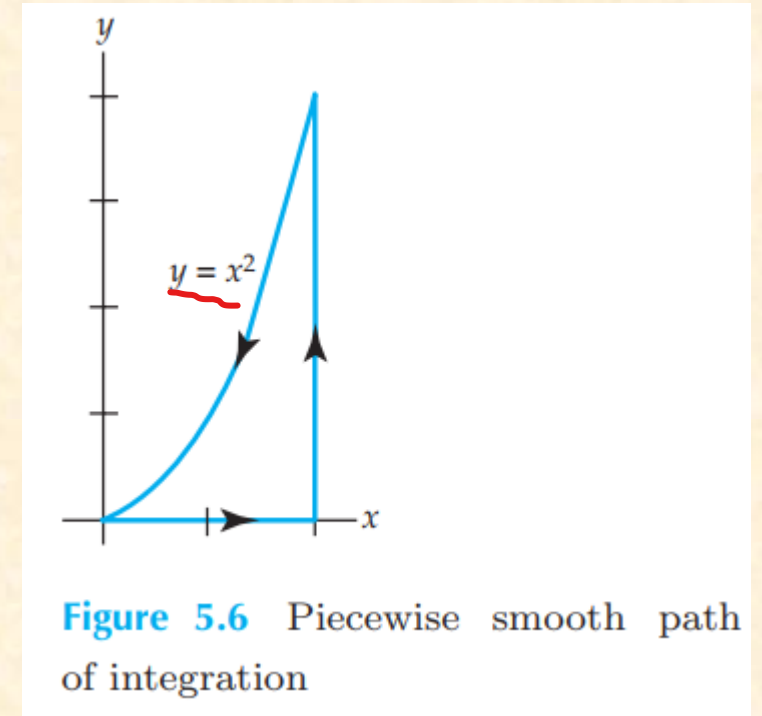
Solution The curve C is illustrated in Figure 5.5 and is defined by the explicit function $y = x^3$. Hence we can use x as the parameter. Using the differential $dy = 3x^2 dx$, we apply (9) and (10):

$$\begin{aligned}\int_C xy dx + x^2 dy &= \int_{-1}^2 x (x^3) dx + x^2 (3x^2 dx) \\ &= \int_{-1}^2 4x^4 dx = \left. \frac{4}{5} x^5 \right|_{-1}^2 = \frac{132}{5}.\end{aligned}$$



Line Integral

- Self Study:** 1. Evaluate $\oint_C x dx$, where C is the circle defined by $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$.
2. Evaluate $\oint_C y^2 dx - x^2 dy$, where C is the closed curve shown in Figure 5.6.

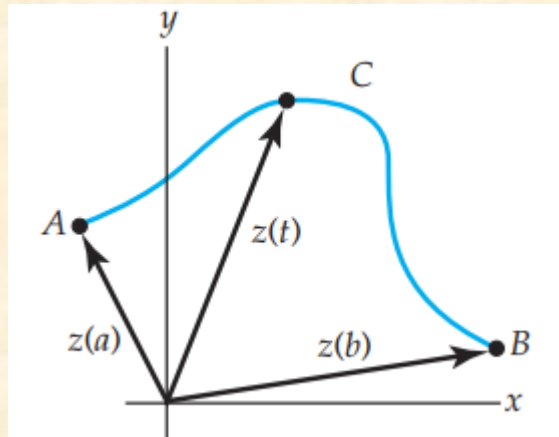


Some Terminologies

Suppose the continuous real-valued functions $x = x(t), y = y(t)$, $a \leq t \leq b$, are parametric equations of a curve C in the complex plane. we can describe the points z on C by means of a complex-valued function of a real variable t called a **parametrization** of C :

$$z(t) = x(t) + iy(t), a \leq t \leq b. \quad (1)$$

The point $z(a) = x(a) + iy(a)$ or $A = (x(a), y(a))$ is called the **initial point** of C and $z(b) = x(b) + iy(b)$ or $B = (x(b), y(b))$ is its **terminal point**.



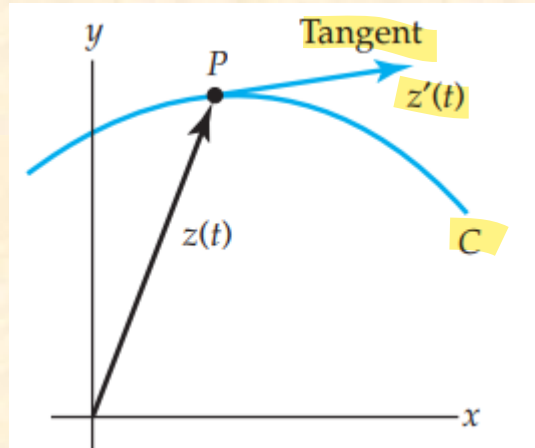
$z(t) = x(t) + iy(t)$ as a position vector

Some Terminologies

Suppose, a curve C in the complex plane for a real variable t :

$$z(t) = x(t) + iy(t), a \leq t \leq b. \quad (1)$$

Suppose the derivative of (1) is $z'(t) = x'(t) + iy'(t)$. We say a curve C in the complex plane is **smooth** if $z'(t)$ is continuous and **never zero in the interval $a \leq t \leq b$** . As shown in the following Figure, since the vector $z'(t)$ is not zero at any point P on C , the vector $z'(t)$ is tangent to C at P . In other words, a smooth curve has a continuously turning tangent;



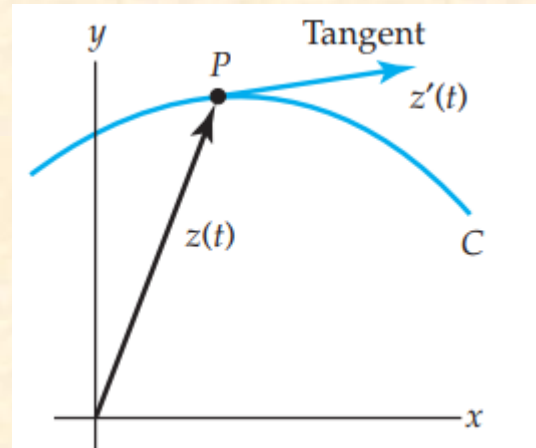
$Z'(t) = x'(t) + iy'(t)$ as a tangent vector

Some Terminologies

Suppose, a curve C in the complex plane for a real variable t :

$$z(t) = x(t) + iy(t), a \leq t \leq b. \quad (1)$$

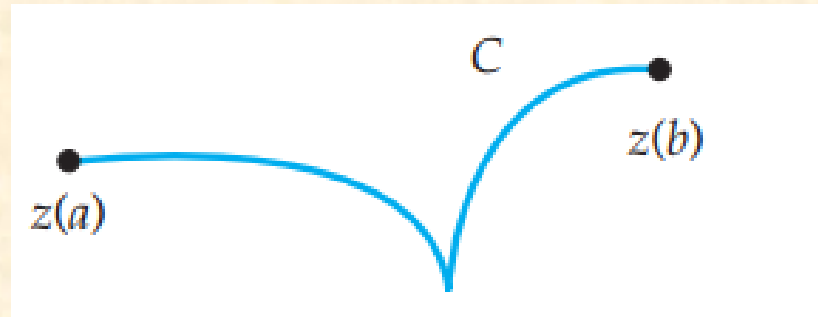
Suppose the derivative of (1) is $z'(t) = x'(t) + iy'(t)$. We say a curve C in the complex plane is **smooth** if $z'(t)$ is continuous and never zero in the interval $a \leq t \leq b$. As shown in the following Figure, since the vector $z'(t)$ is not zero at any point P on C , the vector $z'(t)$ is tangent to C at P . In other words, a smooth curve has a continuously turning tangent;



$Z'(t) = x'(t) + iy'(t)$ as a tangent vector

Some Terminologies

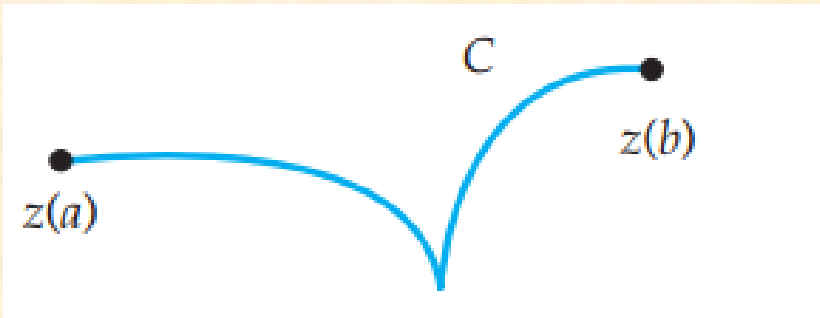
A **piecewise smooth curve** C has a continuously turning tangent, except possibly at the points where the component smooth curves $C_1, C_2, \dots, C_n$ are joined together. a smooth curve can have no sharp corners or cusps. See the following Figure



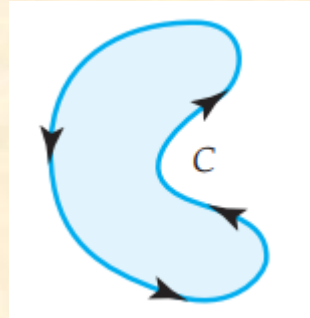
Curve C is not smooth since it has a cusp.

Some Terminologies

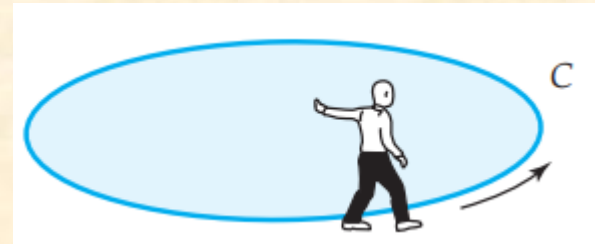
A curve C in the complex plane is said to be a **simple** if $z(t_1) \neq z(t_2)$ for $t_1 \neq t_2$, except possibly for $t = a$ and $t = b$. C is a **closed curve** if $z(a) = z(b)$. C is a **simple closed curve** if $z(t_1) \neq z(t_2)$ for $t_1 \neq t_2$ and $z(a) = z(b)$. In complex analysis, a piecewise smooth curve C is called a **contour** or **path**.



Simple Curve



Simple Closed Curve



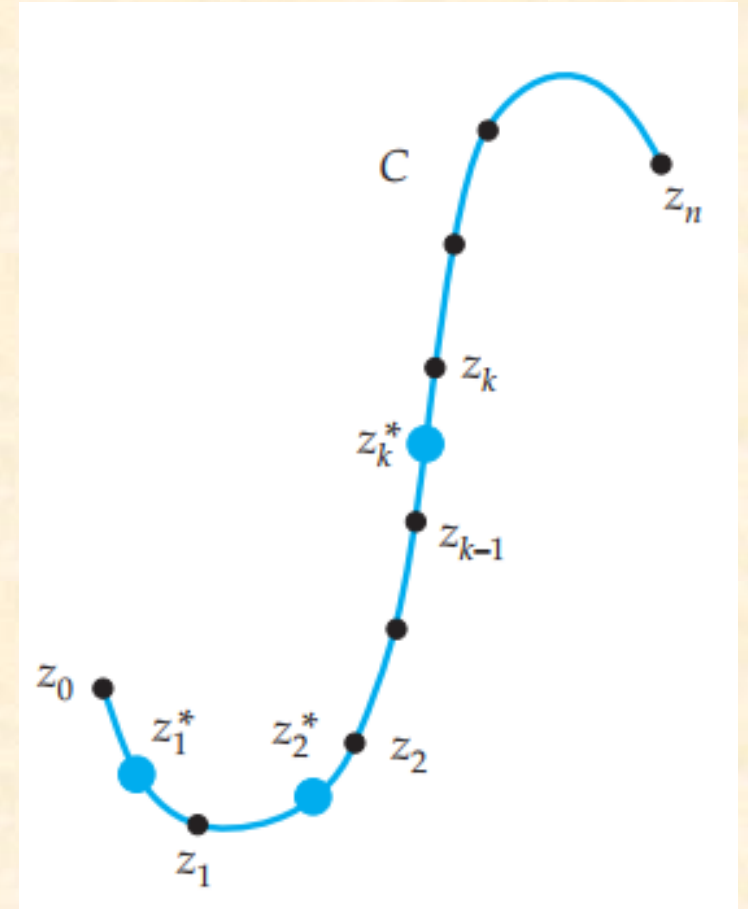
Complex Integration

The complex integration of f on C

$$\int_C f(z) dz = \lim_{\|p\| \rightarrow 0} \sum_{k=1}^n f(z_k^*) \Delta z_k^*$$

If C is a closed curve

$$\oint_C f(z) dz = \lim_{\|p\| \rightarrow 0} \sum_{k=1}^n f(z_k^*) \Delta z_k^*$$



Partition of curve C into n subarcs is induced by a partition P of the parameter interval $[a, b]$.

Theorem: Evaluation of a Contour Integral

If f is continuous on a smooth curve C given by the parametrization $z(t) = x(t) + iy(t)$, $a \leq t \leq b$, then

$$\int_C \underline{f(z)dz} = \int \underline{f(z(t))z'(t) dt}$$

EXAMPLE 1: Evaluating a Contour Integral

Evaluate $\int_C \bar{z} dz$, where C is given by $x = 3t$, $y = t^2$, $-1 \leq t \leq 4$.

Solution From (1) a parametrization of the contour C is $z(t) = 3t + it^2$. Therefore, with the identification $f(z) = \bar{z}$ we have $\underline{f(z(t))} = \overline{3t + it^2} = 3t - it^2$. Also, $z'(t) = 3 + 2it$, and so by (11) the integral is

$$\int_C \underline{\bar{z} dz} = \int_{-1}^4 \underline{(3t - it^2)(3 + 2it) dt} = \int_{-1}^4 \underline{[2t^3 + 9t + 3t^2i] dt}.$$

Theorem: Evaluation of a Contour Integral

EXAMPLE 1: Evaluating a Contour Integral

Now in view of (4), the last integral is the same as

$$\begin{aligned}\int_C \bar{z} dz &= \int_{-1}^4 (2t^3 + 9t) dt + \underline{i} \int_{-1}^4 \underline{3t^2} dt \\ &= \left(\frac{1}{2}t^4 + \frac{9}{2}t^2 \right) \Big|_{-1}^4 + it^3 \Big|_{-1}^4 = 195 + 65i.\end{aligned}$$

Theorem: Evaluation of a Contour Integral

EXAMPLE 2: Evaluating a Contour Integral

Evaluate $\oint_C \frac{1}{z} dz$, where C is the circle $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$.

Solution In this case $z(t) = \cos t + i \sin t = e^{it}$, $z'(t) = ie^{it}$, and $f(z(t)) = \frac{1}{z(t)} = e^{-it}$. Hence,

$$\oint_C \frac{1}{z} dz = \int_0^{2\pi} (e^{-it}) ie^{it} dt = i \int_0^{2\pi} dt = 2\pi i.$$

Theorem: Properties of Contour Integrals

Suppose the functions f and g are continuous in a domain D , and C is a smooth curve lying entirely in D . Then

- (i) $\int_C k f(z) dz = k \int_C f(z) dz$, k a complex constant.
- (ii) $\int_C [f(z) + g(z)] dz = \int_C f(z) dz + \int_C g(z) dz$.
- (iii) $\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz$, where C consists of the smooth curves C_1 and C_2 joined end to end.
- (iv) $\int_{-C} f(z) dz = -\int_C f(z) dz$, where $-C$ denotes the curve having the opposite orientation of C .

Theorem: Evaluation of a Contour Integral

EXAMPLE 3: C Is a Piecewise Smooth Curve

Evaluate $\int_C (x^2 + iy^2) dz$, where C is the contour shown in Figure 5.20.

Solution In view of Theorem 5.2(iii) we write

$$\int_C (x^2 + iy^2) dz = \int_{C_1} (x^2 + iy^2) dz + \int_{C_2} (x^2 + iy^2) dz.$$

Since the curve C_1 is defined by $y = x$, it makes sense to use x as a parameter. Therefore, $z(x) = x + ix$, $z'(x) = 1 + i$, $f(z) = x^2 + iy^2$, $f(z(x)) = x^2 + ix^2$, and

$$\begin{aligned} \int_{C_1} (x^2 + iy^2) dz &= \int_0^1 \overbrace{(x^2 + ix^2)}^{(1+i)x^2} (1+i) dx \\ &= (1+i)^2 \int_0^1 x^2 dx = \frac{(1+i)^2}{3} = \frac{2}{3}i. \end{aligned} \quad (12)$$

Theorem: Evaluation of a Contour Integral

EXAMPLE 3: C Is a Piecewise Smooth Curve

The curve C_2 is defined by $x = 1$, $1 \leq y \leq 2$. If we use y as a parameter, then $z(y) = 1 + iy$, $z'(y) = i$, $f(z(y)) = 1 + iy^2$, and

$$\int_{C_2} (x^2 + iy^2) dz = \int_1^2 (1 + iy^2)i dy = -\int_1^2 y^2 dy + i \int_1^2 dy = -\frac{7}{3} + i. \quad (13)$$

Combining (10) and (13) gives $\int_C (x^2 + iy^2) dz = \frac{2}{3}i + (-\frac{7}{3} + i) = -\frac{7}{3} + \frac{5}{3}i$.

Evaluation of a Contour Integral

Self Study: Evaluate the contour integral of $\int_C f(z)dz$ using the parametric representations for C , where $f(z) = \frac{z^2 - 1}{z}$

and the curve C is

- (a) the semicircle $z = 2e^{i\theta}$ ($0 \leq \theta \leq \pi$);
- (b) the semicircle $z = 2e^{i\theta}$ ($\pi \leq \theta \leq 2\pi$);
- (c) the circle $z = 2e^{i\theta}$ ($0 \leq \theta \leq 2\pi$);

Thanks a lot ...